

Deterministic Pseudorandom Invariants — Math-Clean Formalization (v2)

Abstract

We formalize deterministic pseudorandom invariants (DPIs) as algebraic objects built on finite rings with rigorously certified statistical properties. The framework separates: (i) an algebraic layer of commuting central idempotents that organizes invariants, (ii) a probabilistic layer that certifies near-uniformity via harmonic analysis on finite abelian groups, and (iii) composition rules that control cross-register dependence by an explicit mutual-information penalty. All claims are stated with precise hypotheses and self-contained proofs.

1. Setting and notation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For a positive integer m , write $\mathbb{Z}_m$ for the additive cyclic group and denote its characters by

$$\chi_k(x) = \exp\left(\frac{2\pi i}{m} kx\right), \quad k \in \{0, 1, \dots, m-1\}.$$

A **register** at modulus m is a stochastic process $X^{(m)} = (X_t^{(m)})_{t \geq 1}$ taking values in $\mathbb{Z}_m$.

Throughout, independence statements are interpreted as **conditional independence given a seed** σ unless noted otherwise. That is, random variables $Y_1, \dots, Y_n$ are conditionally independent given σ and we analyze laws and expectations under $\mathbb{P}(\cdot \mid \sigma)$. When no seed is present, read these as ordinary independence.

We write $\text{Law}(Z)$ for the distribution of Z ; U_m for the uniform law on $\mathbb{Z}_m$; and $\|\cdot\|_{\text{TV}}$ for total variation distance.

2. Encoders, compressors, and DPIs

Let $m \geq 2$ and $n \geq 1$. Given n inputs $Y_1, \dots, Y_n \in \mathbb{Z}_m$, a **linear compressor** at modulus m with weight vector $w = (w_1, \dots, w_n) \in (\mathbb{Z}_m)^n$ is

$$H_m(Y_1, \dots, Y_n) := \sum_{i=1}^n w_i Y_i \bmod m.$$

We require weights w_i to be units in $\mathbb{Z}_m$ (i.e., $\gcd(w_i, m) = 1$) to avoid trivial character collapse under pushforward.

A **DPI** is a triple $(\mathcal{A}, \mathcal{P}, \mathcal{H})$ where - $\mathcal{A}$ is a unital ring generated by commuting central idempotents $\{P_j\}_{j=1}^r$ with $\sum_j P_j = 1$, - $\mathcal{P}$ is a family of register processes valued in $\bigoplus_j P_j \mathbb{Z}_{m_j}$, and - $\mathcal{H}$ is a collection of compressors H_{m_j} acting componentwise on the corresponding registers.

The algebraic layer induces a Karoubi-style refinement order on invariants: a refinement applies a further central idempotent decomposition and then compresses within components.

3. Harmonic analysis and a TV bound at modulus m

For a $\mathbb{Z}_m$ -valued random variable Z , its (discrete) Fourier transform is $\hat{\mu}_Z(k) = \mathbb{E}[\chi_k(Z)]$. Parseval and inversion hold in the usual form. We use the following standard inequality.

Lemma 3.1 (Fourier-TV upper bound). For any $\mathbb{Z}_m$ -valued Z ,

$$\|\text{Law}(Z) - U_m\|_{\text{TV}} \leq \frac{1}{2} \sum_{k=1}^{m-1} |\hat{\mu}_Z(k)|.$$

Proof. By orthogonality of characters and the usual Fourier expansion of signed measures, see any text on harmonic analysis on finite abelian groups. $\square$

Lemma 3.2 (Product structure for linear sums). Let $Y_1, \dots, Y_n \in \mathbb{Z}_m$ be conditionally independent given σ . For weights $w \in (\mathbb{Z}_m^\times)^n$,

$$\hat{\mu}_{\sum_i w_i Y_i | \sigma}(k) = \prod_{i=1}^n \hat{\mu}_{Y_i | \sigma}(kw_i) \quad (k \in \mathbb{Z}_m).$$

Proof. Independence yields factorization of conditional characteristic functions; linearity sends $\chi_k(\sum_i w_i Y_i) = \prod_i \chi_{kw_i}(Y_i)$. $\square$

Corollary 3.3 (Compressor TV bound). Under the conditions of Lemma 3.2,

$$\|\text{Law}(H_m(Y_1, \dots, Y_n)) - U_m\|_{\text{TV}} \leq \frac{1}{2} \sum_{k=1}^{m-1} \prod_{i=1}^n |\hat{\mu}_{Y_i}(kw_i)| \leq \frac{1}{2} (m-1) \prod_{i=1}^n \rho_i,$$

where $\rho_i := \max_{k \in \{1, \dots, m-1\}} |\hat{\mu}_{Y_i}(k)|$. If each $\rho_i \leq \rho < 1$ then $\|\cdot\|_{\text{TV}} \leq \frac{1}{2} (m-1) \rho^n$.

Remarks. (1) Unit weights prevent a nontrivial $k \neq 0$ from mapping to $kw_i \equiv 0 \pmod{m}$, which would otherwise force $|\hat{\mu}_{Y_i}(kw_i)| = 1$ and weaken the bound. (2) The bound is tight enough for schedule design: exponential decay in n once $\rho < 1$.

4. Cross-register composition with mutual information control

Consider two moduli m_1, m_2 and random variables $Z_1 \in \mathbb{Z}_{m_1}$ and $Z_2 \in \mathbb{Z}_{m_2}$. Let $I(Z_1; Z_2)$ denote mutual information.

Theorem 4.1 (Bivariate composition).

$$\|\text{Law}(Z_1, Z_2) - U_{m_1} \times U_{m_2}\|_{\text{TV}} \leq \|\text{Law}(Z_1) - U_{m_1}\|_{\text{TV}} + \|\text{Law}(Z_2) - U_{m_2}\|_{\text{TV}} + \sqrt{\frac{1}{2} I(Z_1; Z_2)}.$$

Proof. Triangle inequality:

$$\|P_{Z_1 Z_2} - U_{m_1} \times U_{m_2}\| \leq \|P_{Z_1 Z_2} - P_{Z_1} \times P_{Z_2}\| + \|P_{Z_1} \times P_{Z_2} - U_{m_1} \times P_{Z_2}\| + \|U_{m_1} \times P_{Z_2} - U_{m_1} \times U_{m_2}\|.$$

The middle terms equal $\|P_{Z_1} - U_{m_1}\|$ and $\|P_{Z_2} - U_{m_2}\|$. Pinsker's inequality yields $\|P_{Z_1 Z_2} - P_{Z_1} \times P_{Z_2}\| \leq \sqrt{\frac{1}{2} \text{KL}(P_{Z_1 Z_2} \| P_{Z_1} \times P_{Z_2})} = \sqrt{\frac{1}{2} I(Z_1; Z_2)}$. $\square$

Corollary 4.2 (48\-, \-, 256 interface). With $m_1 = 48$ and $m_2 = 256$, the joint deviation from uniform is at most the sum of the marginals' deviations and the information penalty $\sqrt{\frac{1}{2} I(Z_{48}; Z_{256})}$.

5. Algebraic layer: central idempotents and refinement

Let $\mathcal{A}$ be a unital ring over which the register family acts, and suppose there exist commuting central idempotents $P_1, \dots, P_r \in \mathcal{A}$ with $\sum_j P_j = 1$.

Theorem 5.1 (Decomposition). The module generated by the DPI splits as a direct sum $M \cong \bigoplus_{j=1}^r P_j M$. Invariants computed after compression respect this decomposition.

Proof. Central idempotents are orthogonal projectors with $P_i P_j = 0$ ($i \neq j$) and $\sum_j P_j = 1$. Standard ring-module decomposition yields the direct sum. Compression acts componentwise by construction. $\square$

Theorem 5.2 (Refinement stability). If $\{Q_{j\ell}\}_\ell$ is for each j a family of commuting central idempotents summing to P_j , then the DPI obtained by replacing P_j with $\{Q_{j\ell}\}_\ell$ produces invariants equal to those from first compressing under P_j and then refining within P_j .

Proof. Functoriality of central idempotent splitting and commutation with linear compressors. $\square$

6. Receipt templates for establishing $\rho < 1$

To apply Corollary 3.3 one needs explicit bounds on per-index spectral radii ρ_i . The following **receipt templates** instantiate $\rho_i \leq \rho < 1$ under checkable conditions. Each receipt is meant to be validated on the concrete generator before acceptance.

1. **Spectral-gap Markov receipt.** Suppose $(Y_{i,t})_{t \geq 1}$ is a time-homogeneous Markov chain on $\mathbb{Z}_m$ with stationary law U_m and L^2 spectral gap $\gamma \in (0, 1]$. Then for each nontrivial character, $|\mathbb{E} \chi_k(Y_{i,t})| \leq (1 - \gamma)^t$. Set sampling stride $t \geq t_*$ with $(1 - \gamma)^{t_*} \leq \rho$.
2. **Exponential-sum receipt (polynomial phases).** Let $Y_i = P_i(T) \bmod m$ for a polynomial P_i with coefficients not all divisible by prime factors of m . For each $k \not\equiv 0 \pmod{m}$, classical bounds on complete exponential sums imply $|\frac{1}{N} \sum_{t=1}^N \chi_k(P_i(t))| \leq C(m, \deg P_i) N^{-1/2}$. Choose block length N so that the bound is $\leq \rho$.
3. **Hyperbolic toral automorphism receipt.** Let Y_i be the projection modulo m of a mixing automorphism on the 2-torus with eigenvalues $|\lambda| > 1 > |\lambda'|$. Discrete factors inherit exponential correlation decay: $|\widehat{\mu}_{Y_i}(k)| \leq Cr^t$ after t steps. Pick stride so $Cr^t \leq \rho$.
4. **Piecewise expanding map receipt (Lasota–Yorke).** If Y_i arises from a piecewise C^2 expanding map with bounded distortion and a Gibbs–Markov partition, its transfer operator has spectral radius < 1 on BV. This yields $|\widehat{\mu}_{Y_i}(k)| \leq Cr^t$ for nontrivial characters after t steps. Choose t with $Cr^t \leq \rho$.
5. **Finite-sample receipt (DKW).** For IID draws with cdf F on $\mathbb{Z}_m$, with N samples the empirical law $\widehat{F}$ satisfies $\mathbb{P}\{\|\widehat{F} - F\|_\infty > \varepsilon\} \leq 2 \exp(-2N\varepsilon^2)$. Combine with a bound on $\max_k |\widehat{\mu}_Y(k)|$ estimated from samples to certify ρ at a target confidence.

Any of these receipts, instantiated per index, deliver the uniform bound $\rho_i \leq \rho < 1$ required by Corollary 3.3.

7. Schedule design and targets

Fix a modulus m and choose n inputs with a common spectral envelope $\rho < 1$. Then

$$\|\text{Law}(H_m(Y_1, \dots, Y_n)) - U_m\|_{\text{TV}} \leq \frac{1}{2}(m-1)\rho^n.$$

Given a TV target $\tau_m \in (0, 1)$, take

$$n \geq \frac{\log(2\tau_m/(m-1))}{\log \rho}.$$

For a pair of registers at moduli 48 and 256 with targets τ_{48}, τ_{256} , Theorem 4.1 yields a joint target $\tau_\times$ provided

$$\sqrt{\frac{1}{2}I(Z_{48}; Z_{256})} \leq \tau_\times - \tau_{48} - \tau_{256}.$$

This sets an explicit quantitative goal on the cross-register mutual information.

8. Computability and resource model

A DPI pipeline is **deterministic budget-0** when it satisfies: 1. Deterministic arithmetic over $\mathbb{Z}_m$ with a fixed word-size instruction set. 2. No calls to randomness or external oracles. 3. Per-tick cost bounded by a modulus-dependent constant, independent of data values and time. 4. Compressors H_m use only additions and multiplications modulo m , with $w_i \in \mathbb{Z}_m^\times$.

Under these rules, each DPI step has constant amortized cost and is reproducible.

9. DPI acceptance checks

For a concrete instance, acceptance requires logged evidence of: 1. **Independence**. Statement and verification that inputs $\{Y_i\}$ are independent or conditionally independent given the specified seed. 2. **Spectral receipts**. A bound $\rho_i \leq \rho < 1$ per input via one of the templates in §6, with parameters and confidence. 3. **Compressor design**. Unit weights w_i at each modulus and the chosen n satisfying the inequality in §7. 4. **Cross-register coupling**. An empirical mutual-information estimate and an upper bound meeting Theorem 4.1's requirement for the intended joint target. 5. **Determinism**. Evidence that the implementation meets §8.

10. Proof supplements

Proof of Lemma 3.1 (detail). Let $\nu = \text{Law}(Z) - U_m$. Then ν has total mass zero and Fourier coefficients $\hat{\nu}(k) = \hat{\mu}_Z(k)$ for $k \neq 0$ and 0 at $k = 0$. The inversion formula gives $\nu(x) = \frac{1}{m} \sum_{k=1}^{m-1} \hat{\mu}_Z(k) \overline{\chi_k(x)}$. Hence

$$\|\nu\|_{\text{TV}} = \frac{1}{2} \sum_x |\nu(x)| \leq \frac{1}{2} \sum_x \frac{1}{m} \sum_{k=1}^{m-1} |\hat{\mu}_Z(k)| = \frac{1}{2} \sum_{k=1}^{m-1} |\hat{\mu}_Z(k)|.$$

Proof of Corollary 3.3 (detail). Combine Lemma 3.1 with Lemma 3.2 and apply the bound $\max_k a_k \leq \frac{1}{m-1} \sum_k a_k$.

Pinsker constant. We use $\|P - Q\|_{\text{TV}} \leq \sqrt{\frac{1}{2} \text{KL}(P\|Q)}$. Any equivalent constant choice can be substituted; change the target accordingly.

11. Practical checklist (per modulus m)

- Specify inputs $Y_1, \dots, Y_n$ and the independence mode.

- Choose unit weights w and compressor size n .
 - Provide a receipt instantiating ρ with numeric parameters.
 - Compute the TV upper bound from §7.
 - If composing registers, measure $I(\cdot; \cdot)$ and verify Theorem 4.1.
 - Log determinism and constant-time compliance.
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12. Summary

The DPI framework is valid under explicit, minimal hypotheses. Linear compression at modulus m admits a clean Fourier-based TV bound under (conditional) independence. Cross-register composition is controlled by a sharp triangle-Pinsker inequality. The algebraic layer of commuting central idempotents remains stable under refinement and does not interfere with the probabilistic guarantees. Receipt templates give concrete, auditable paths to certification.